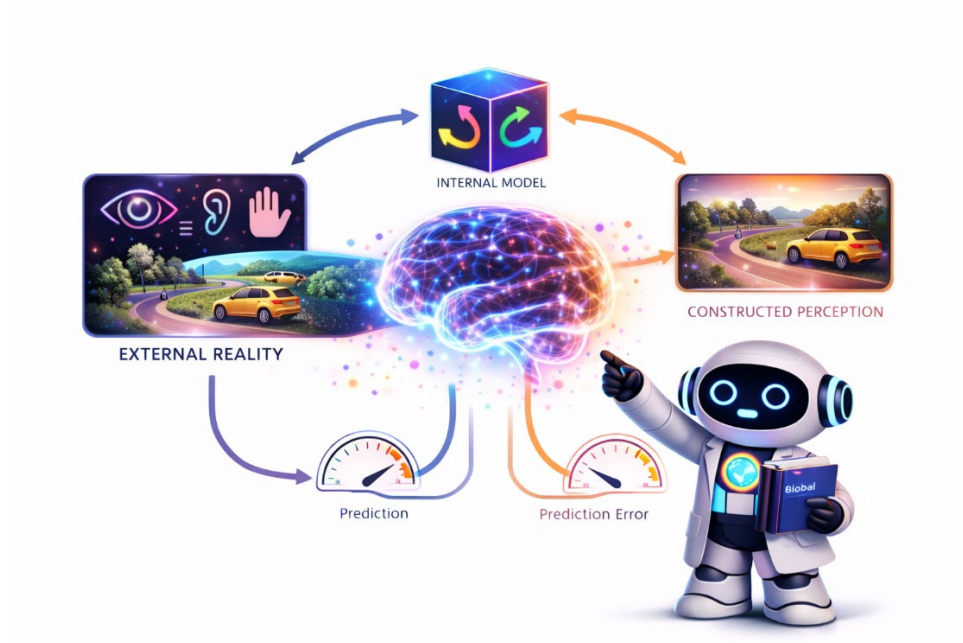


FOUNDATIONS of NEUROSCIENCE & THE MODEL OF THE WORLD

How the Brain Constructs Reality



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Neuroscience as a Model for AI (ITAI 4374)

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Section 1: Where We Are

Before we dive into new material, let's take a moment to orient ourselves. Where are we in this course? What did we learn last time? And where are we headed?

1.1 Your Journey Through Neuroscience for AI



FIGURE 1.1: Course Journey Roadmap

This course takes you on a journey from asking 'Why should AI practitioners care about the brain?' to understanding specific brain mechanisms that could inspire better AI systems.

YOUR POSITION IN THE COURSE

Module 01 (Completed): Why Neuroscience for AI? — The case for brain-inspired AI

Module 02 (Today): Foundations & Models of the World — What makes brains special?

Module 03 (Next): Brain Anatomy & Functions — The parts and what they do

Module 04 (Coming): Neurons & Neural Computation — How brain cells process information

1.2 Module 01 Recap: Key Takeaways

Let's refresh what we learned in Module 01:

The Historical Connection

AI and neuroscience have always been intertwined. The perceptron was explicitly modeled on neurons. The term 'deep learning' refers to the brain's hierarchical processing. Neural networks are literally named after the brain's computational units.

The Divergence

For decades, the fields diverged. AI researchers optimized for benchmarks and computational efficiency, often abandoning biological accuracy. The mantra became 'whatever works'—and it worked spectacularly well, giving us systems like GPT, AlphaFold, and DALL-E.

The Reconvergence

But now something interesting is happening. Current AI, despite its power, has hit certain walls that brains don't seem to have: Learning from few examples. Energy efficiency. Common sense reasoning. Knowing what it doesn't know. These aren't minor gaps—they're fundamental. And they're exactly the things brains do well.

Your Unique Position

As AI practitioners, you already know modern AI inside and out—ANNs, deep learning, computer vision, NLP, generative AI, LLMs. Now we add a new lens: the biological one. Not to replace what you know, but to complement it with insights from the only system we know of that achieves general intelligence.

DID YOU KNOW?

The term 'artificial neural network' was chosen specifically because early researchers were trying to model the brain. Frank Rosenblatt's perceptron (1958) was designed as a simplified model of how neurons might learn. The connection between AI and neuroscience isn't just historical—it's foundational.

1.3 The Bridge to Module 02

Module 01 asked WHY we should study the brain. Module 02 asks WHAT the brain does that's so remarkable.

Think of it this way: Last time we made the case that the brain has something to teach us. This time we start learning what that something is. We'll explore the central concept of 'world models'—the idea that the brain actively constructs reality rather than passively receiving it—and examine four major theories about how it accomplishes this feat.

By the end of this module, you'll have a framework for thinking about brain function that will inform everything that follows.

Section 2: What is Neuroscience?

Many of you come from computer science, engineering, or mathematics backgrounds. You know algorithms and data structures intimately, but you may never have formally studied the brain. So let's start with the basics: What exactly is neuroscience, and why should you care?

2.1 Defining the Field

Textbook Definition: Neuroscience is the scientific study of the nervous system—the brain, spinal cord, and neural networks throughout the body.

Practical Definition: Neuroscience is the attempt to understand how three pounds of biological tissue creates your entire experience of being alive.

Think about what your brain is doing right now. It's converting patterns of light into the experience of reading. It's extracting meaning from arbitrary squiggles on a page. It's regulating your heartbeat, monitoring for threats, maintaining your body temperature, and processing a constant stream of sensory information, all while consuming about 20 watts of power. That's less than a dim light bulb.

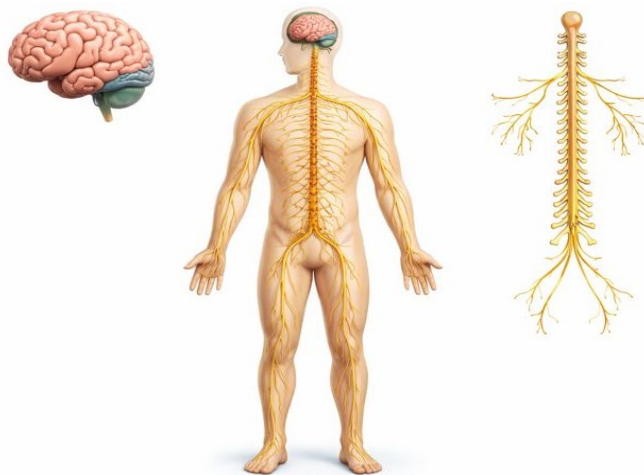


FIGURE 2.1: The Scope of Neuroscience

BY THE NUMBERS: THE BRAIN'S VITAL STATISTICS

Weight: ~1.4 kg (about 3 pounds)

Neurons: ~86 billion

Synaptic connections: ~100 trillion

Power consumption: ~20 watts

Percentage of body's energy: ~20% (despite being only 2% of body weight)

Processing speed: Neurons fire at ~100 Hz (compare to GHz processors)

2.2 The Subfields of Neuroscience

Neuroscience isn't one field: it's many fields united by a common subject. Understanding these subfields helps you know which type of neuroscience is most relevant for AI.

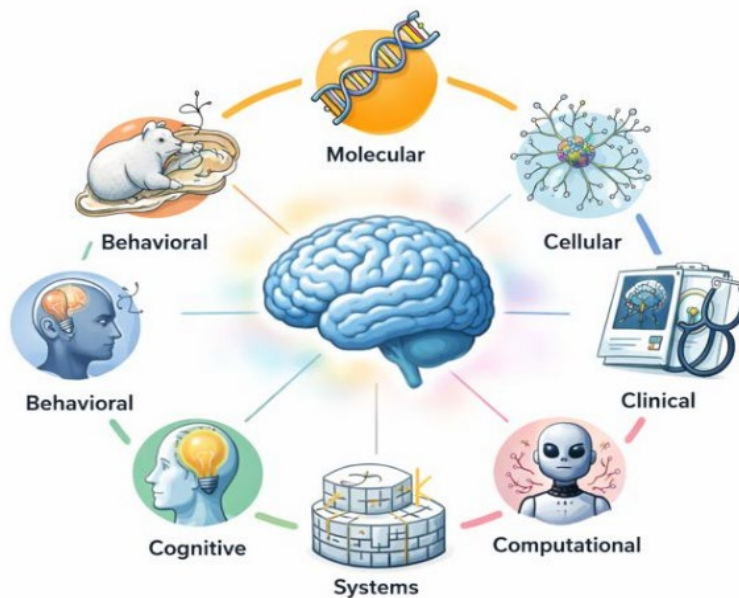


FIGURE 2.2: Subfields of Neuroscience

Molecular Neuroscience

Studies the genes, proteins, and chemical signals involved in neural function. If you want to understand what molecules make neurons work, this is your field.

Cellular Neuroscience

Focuses on individual neurons—their structure, electrical properties, and how they communicate. This is where you learn about action potentials and synaptic transmission.

Systems Neuroscience

Examines how neurons work together in circuits and brain regions. This field asks questions like 'How does the visual cortex process images?'

Behavioral Neuroscience

Links brain function to behavior. How do different brain states lead to different actions?

Cognitive Neuroscience ★

Studies higher mental functions like memory, attention, language, and decision-making. This is highly relevant for AI because these are exactly the capabilities we want our systems to have.

Computational Neuroscience ★

Uses mathematical models to understand brain function. This field asks 'What algorithms does the brain run?' It's the most directly relevant subfield for AI researchers because it bridges biology and computation.

Clinical Neuroscience

Studies brain disorders and their treatments. While not directly relevant to AI, it provides valuable insights into what happens when specific brain functions fail.

AI CONNECTION

For this course, we focus primarily on Computational and Cognitive Neuroscience. We care less about which specific molecules are involved and more about the computational PRINCIPLES—the algorithms and architectures that make the brain work. It's like how a computer scientist cares about algorithms without needing to understand transistor physics.

2.3 A Brief History of Brain Science

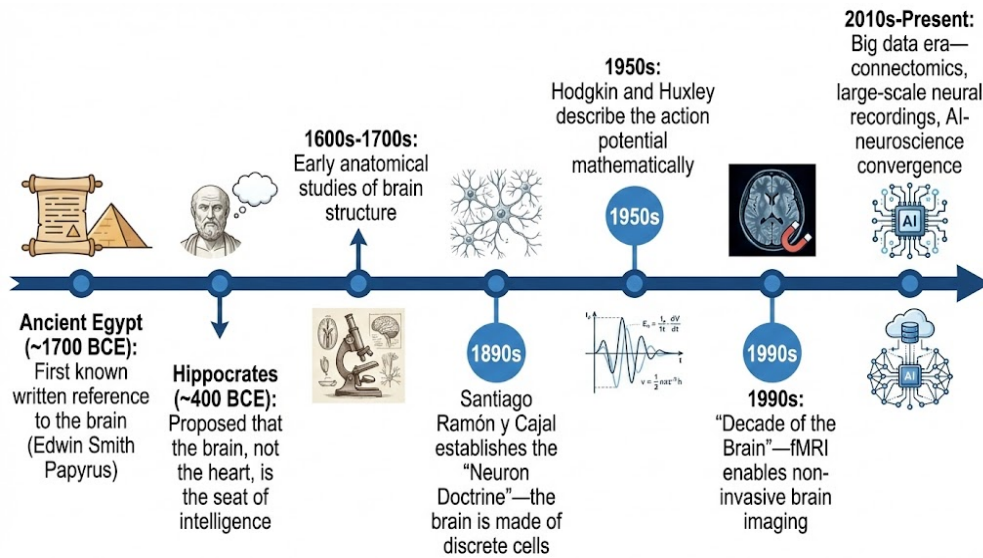


FIGURE 2.3: Timeline of Neuroscience

The study of the brain has a long history:

- Ancient Egypt (~1700 BCE): First known written reference to the brain (Edwin Smith Papyrus)
- Hippocrates (~400 BCE): Proposed that the brain, not the heart, is the seat of intelligence
- 1600s-1700s: Early anatomical studies of brain structure
- 1890s: Santiago Ramón y Cajal establishes the 'Neuron Doctrine'—the brain is made of discrete cells
- 1950s: Hodgkin and Huxley describe the action potential mathematically
- 1990s: 'Decade of the Brain'—fMRI enables non-invasive brain imaging
- 2010s-Present: Big data era—connectomics, large-scale neural recordings, AI-neuroscience convergence

2.4 Why Neuroscience is Hard

Before we proceed, it's worth acknowledging why understanding the brain is so challenging:

The Complexity Challenge

The brain contains roughly 86 billion neurons, each connected to thousands of others. That's about 100 trillion synapses—more than the number of stars in the Milky Way galaxy. And unlike stars, these connections are constantly changing based on experience.

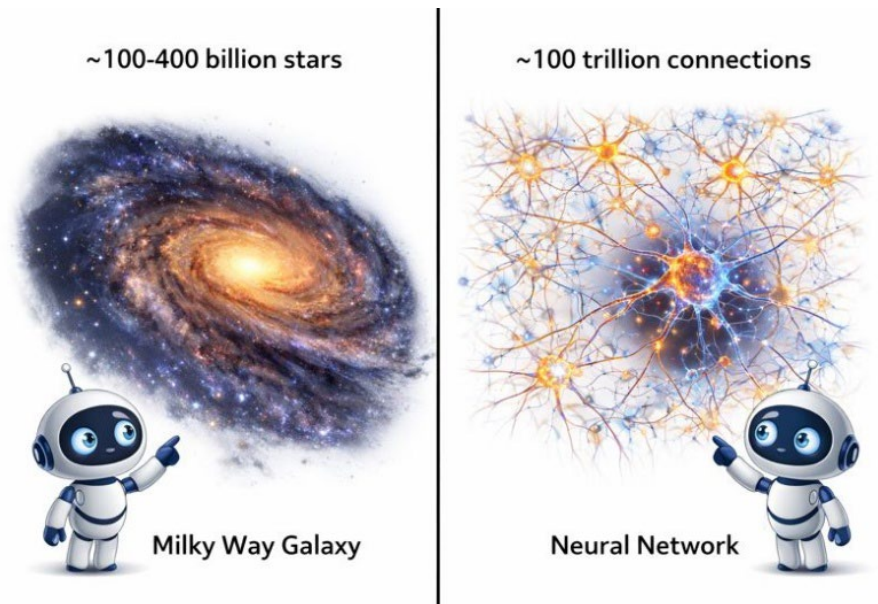


FIGURE 2.4: The Complexity Challenge

The Scale Problem

Brain processes happen at multiple scales simultaneously: molecular (nanometers), cellular (micrometers), circuit (millimeters), regional (centimeters), and whole-brain (decimeters). Understanding any one scale doesn't automatically explain the others.

The Self-Reference Problem

The brain is trying to understand itself. This creates unique philosophical and methodological challenges. As neuroscientist David Eagleman puts it: 'The brain is the most complex object in the known universe, and it's trying to understand itself with a limited budget.'

QUOTEABLE

"All models are wrong, but some are useful." — George Box, statistician

This quote is essential for understanding our approach. The theories we'll explore are all, in some sense, 'wrong'—they're simplifications of an incredibly complex system. But they're useful because they capture important truths about how the brain works and suggest new directions for AI.

Section 3: Why Study the Brain for AI?

We touched on this in Module 01, but let's go deeper. What specifically can the brain teach us that we can't learn from pure engineering approaches?

3.1 Two Paths to Intelligence

There are fundamentally two approaches to building intelligent machines:

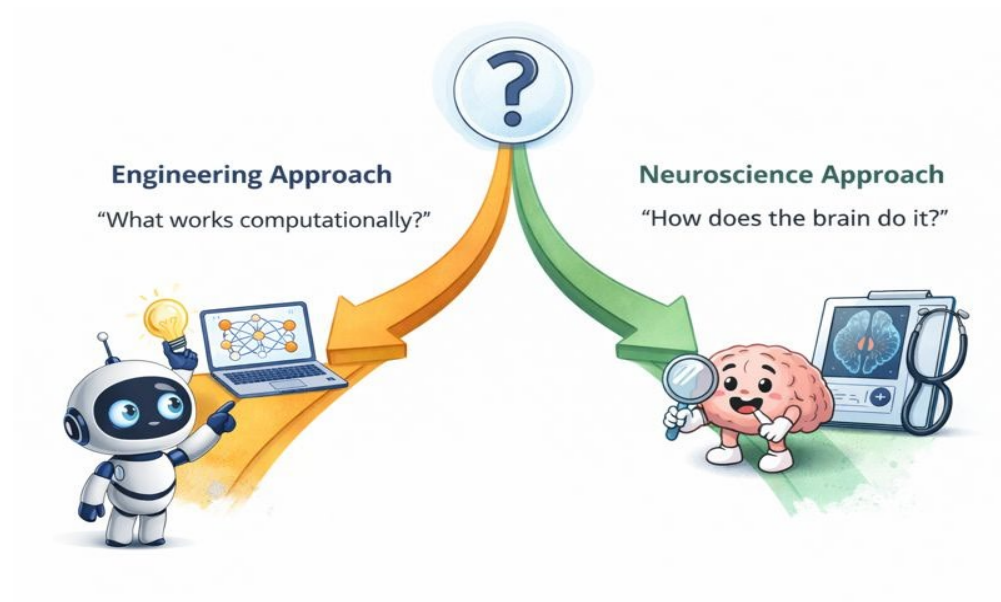


FIGURE 3.1: Two Paths to Intelligence

The Engineering Approach

'What works computationally?' This approach optimizes for performance on benchmarks, using whatever mathematical techniques help—whether or not they resemble anything biological. This has been spectacularly successful, giving us transformers, diffusion models, and reinforcement learning systems that master complex games.

The Neuroscience Approach

'How does the brain do it?' This approach assumes that evolution already solved general intelligence, so we should study that solution. The brain exists and works—it's proof that intelligence is achievable with available resources.

For most of AI's history, these paths diverged. Researchers who stayed close to biology were often left behind by those who just optimized what worked. But now the paths are starting to reconverge. Why?

3.2 The Limitations Gap

Current AI, despite its impressive capabilities, struggles with things that brains do naturally:

Capability	 Brain	 AI
Learning from few examples (one-shot learning)	✓	✗
Energy efficiency (20W vs megawatts)	✓	✗
Common sense reasoning	✓	✗
Adapting to new situations	✓	✗
Knowing what it doesn't know	✓	✗
Continuous lifelong learning	✓	✗

 THE PERFORMANCE PARADOX

Current AI can beat world champions at chess and Go, generate photorealistic images, and write coherent essays—yet a three-year-old outperforms it at learning new concepts from single examples, understanding physical causality, and adapting to novel situations. This gap suggests we're missing something fundamental about how intelligence works.

3.3 The Brain as Proof of Concept

Here's a powerful argument for why neuroscience matters for AI:

1. The brain achieves general intelligence
2. The brain is a physical system
3. Therefore: General intelligence is achievable by physical systems
4. If we understand the principles → we can implement them

The brain is PROOF that human-level intelligence is possible with available resources. It fits in a shoebox. It runs on 20 watts. Its components are slow—neurons fire at maybe 100 Hz, compared to gigahertz processors. Yet it develops from a single cell and achieves general intelligence. This is not speculation—it's empirical fact.



THE AIRPLANE ANALOGY

We didn't build airplanes by copying birds feather-by-feather. But understanding that wings generate lift through airflow was essential. Similarly, we don't need to copy the brain neuron-by-neuron. We need to understand its computational principles. The question is: What are those principles?

3.4 Marr's Three Levels of Analysis

Vision scientist David Marr proposed that any information-processing system can be understood at three levels:

Computational Level

WHAT does the system do and WHY? What problem is it solving? For vision: 'Extract information about objects and surfaces from patterns of light.'

Algorithmic Level

HOW does it solve the problem? What representations and processes does it use? For vision: 'Build hierarchical feature detectors that combine edges into shapes.'

Implementation Level

What PHYSICAL SUBSTRATE carries out the algorithm? For vision: 'Neurons in the visual cortex connected in specific patterns.'

In this course, we focus primarily on the computational and algorithmic levels. We want to understand WHAT the brain computes and HOW, without getting lost in the biological details of implementation. This is exactly what's useful for AI: principles that can be implemented in silicon rather than wetware.

Section 4: The Brain's Greatest Magic Trick

We've now arrived at the central concept of this entire module—perhaps the most important idea in the course.

4.1 Sensation vs. Perception: A Critical Distinction

Before we proceed, let's clarify two terms that are often confused:

Term	Definition	Characteristics
Sensation	The raw transduction of physical energy into neural signals	Relatively faithful to physical reality; happens at sensory receptors
Perception	The brain's constructed interpretation of those signals	Active inference; can be 'wrong' (illusions); incorporates prior knowledge

This distinction explains why perception can be 'wrong' (as in illusions) while the underlying sensory signals are relatively accurate. The brain doesn't just pass along what the senses detect—it interprets, fills in, and constructs.

4.2 The Brain in Darkness

Here is a fact that should astonish you:

Your brain has never seen light.

Really think about that. Your brain sits in complete darkness inside your skull. It has never directly experienced light, sound, touch, or any other sensation. All it receives are electrical signals—billions of them per second—from sensory neurons that translate physical energy into electrochemical impulses.



FIGURE 4.1: The Brain in Darkness

From this chaotic storm of electrical impulses, the brain somehow generates:

- Vivid, full-color visual scenes
- The faces of people you love
- The taste of your favorite food
- Music that moves you to tears
- Your entire experience of being YOU

How? This is what neuroscientists call the brain's 'model of the world.'

THE CENTRAL INSIGHT

The brain doesn't passively receive reality like a camera recording video. It actively **CONSTRUCTS** a model—a simulation—based on sensory input and prior knowledge. What you experience as 'the world out there' is actually a simulation running inside your head.

4.3 What is a 'Model of the World'?

A 'model of the world' (sometimes called a 'world model' or 'internal model') is the brain's internal representation of external reality. It's not a photograph or recording—it's a dynamic, predictive simulation that:

- Represents objects, people, and spaces and their properties
- Encodes relationships between things
- Generates predictions about what will happen next
- Updates based on sensory input
- Allows mental simulation of actions before taking them
- Supports reasoning about counterfactuals ('what if...')

Think of it like this: your brain is constantly running a simulation of reality, and your conscious experience is the output of that simulation. Sensory input doesn't create the experience directly—it updates the simulation.

4.4 Why World Models Matter

For Neuroscience

- Explains why perception is active, not passive
- Explains why illusions work (the model makes 'mistakes')
- Explains imagination and planning (we manipulate the model internally)
- Explains effects of brain damage (parts of the model get disrupted)

For AI

- Highlights what current systems might be missing (pattern matching ≠ understanding)
- Suggests new architectures (systems that build and maintain models)
- Explains certain AI failures (no world model → inconsistent behavior)
- Points toward more robust, generalizable systems

4.5 Brain vs. Current AI: A Comparison

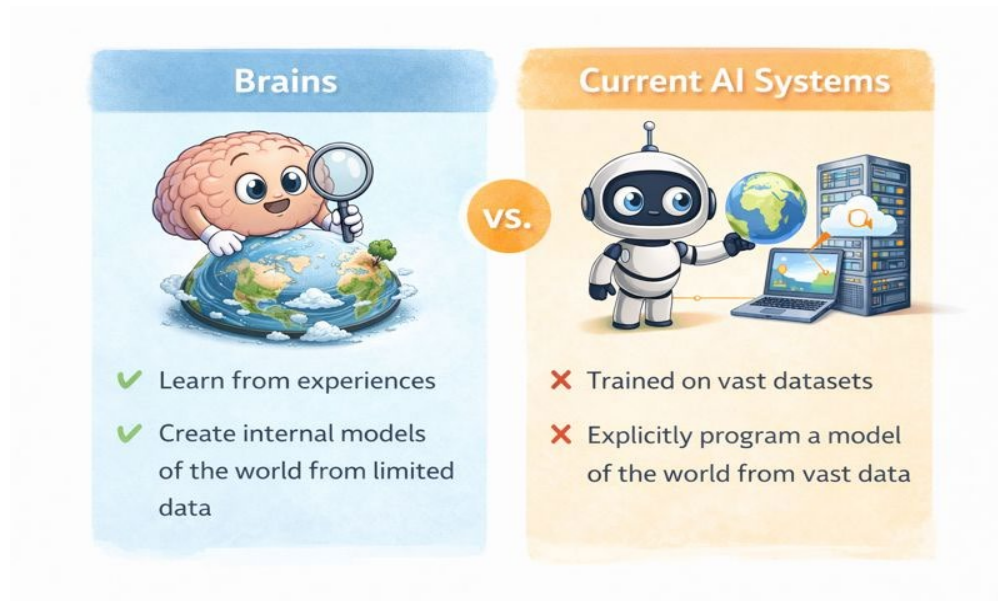


FIGURE 4.2: World Models Comparison

Aspect	 Brain	 Current AI
World Model	Explicit, updatable	Implicit/Emergent (debated)
Prediction	Core function	Task-specific
Uncertainty	Explicit representation	Often overconfident
Imagination	Model manipulation	Interpolation in latent space
Consistency	Unified model	Can be inconsistent

4.6 The Big Question

We've established that the brain builds a model of the world. But HOW does it do this? What are the mechanisms? What's the architecture?

This is where the four theories we'll explore come in. Each offers a different perspective on how the brain constructs and uses its world model:

- **Thousand Brains Theory:** Focuses on REPRESENTATION—how knowledge is structured using reference frames
- **Predictive Coding:** Focuses on PROCESSING—how predictions flow down and errors flow up
- **Bayesian Brain:** Focuses on COMPUTATION—how beliefs are updated given evidence
- **Global Workspace Theory:** Focuses on ACCESS—how information becomes conscious and globally available

These theories aren't mutually exclusive—they may all be true simultaneously, describing different aspects of the same underlying system. Let's explore each one.

Section 5: The Problem of Perception

Before we dive into specific theories, we need to understand WHY the brain needs to construct a model at all. Why can't it just passively receive reality?

5.1 Helmholtz and Unconscious Inference

In the 1860s, German physicist and physician Hermann von Helmholtz made a revolutionary proposal: perception is not passive reception but active inference.

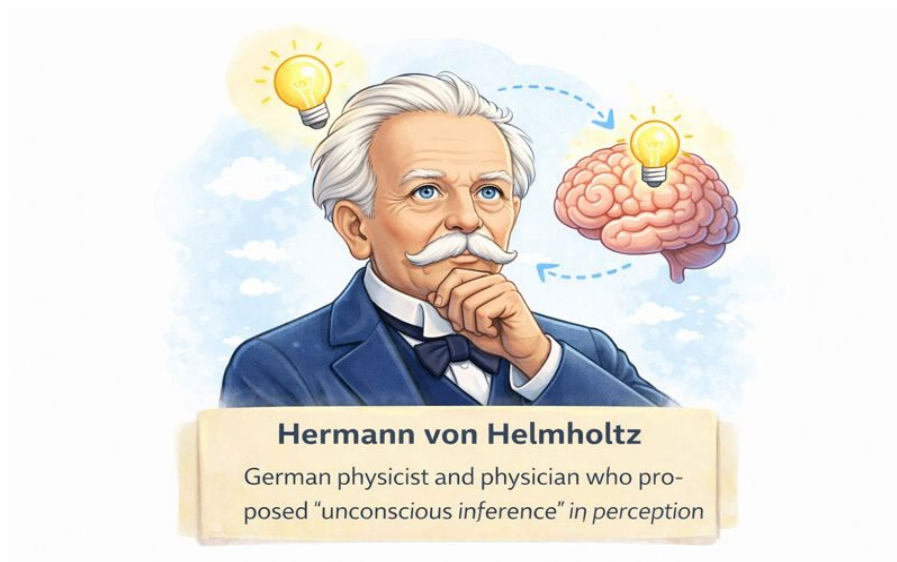


FIGURE 5.1: Hermann von Helmholtz (1821-1894)

Helmholtz argued that sensory data is inherently ambiguous. The patterns of light hitting your retina could have been caused by countless different configurations of objects, surfaces, and lighting. To resolve this ambiguity, the brain must make inferences—educated guesses based on prior experience.

He called this 'unbewusster Schluss' or 'unconscious inference.' The brain is constantly drawing conclusions about the causes of sensory input, and we experience the conclusions, not the raw data.

💬 QUOTABLE

"The psychic activities that lead us to infer that there in front of us at a certain place there is a certain object of a certain character, are generally not conscious activities, but unconscious ones. In their result they are equivalent to a conclusion..."

— Hermann von Helmholtz (1867)

5.2 The Inverse Problem

Helmholtz identified what's now called the 'inverse problem' of perception. Here's the issue:

The Forward Problem (Easy)

Given a 3D scene with objects, surfaces, and lighting, calculate what pattern of light would reach the eye. This is what computer graphics does—it's well-defined and has a unique solution.

The Inverse Problem (Hard)

Given a pattern of light reaching the eye, infer what 3D scene caused it. This is what perception must do—and it has infinitely many solutions!

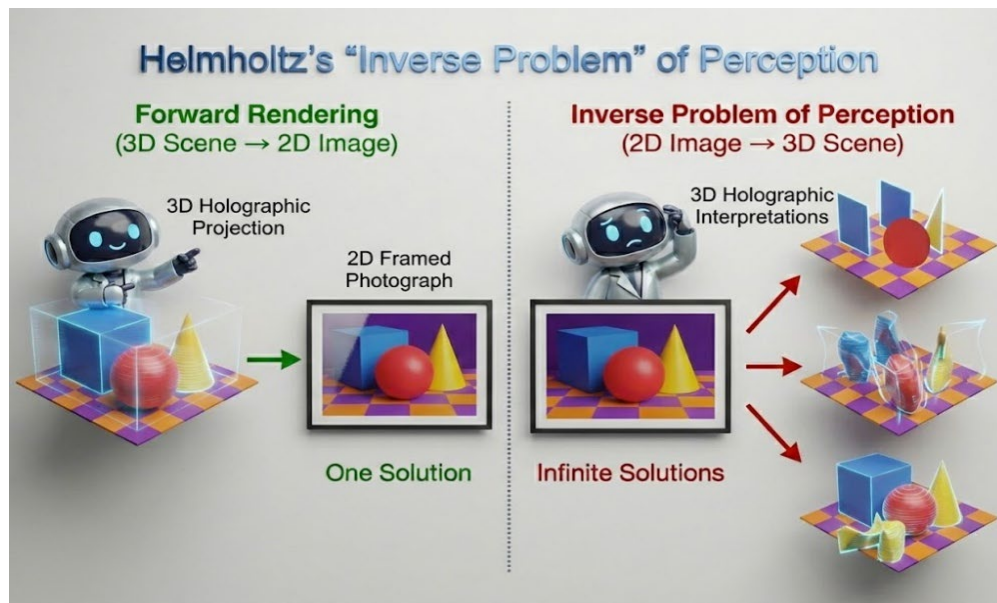


FIGURE 5.2: The Inverse Problem

Think about it: a given 2D retinal image could be produced by a nearby small object or a distant large one. It could be produced by a flat surface with painted shadows or a 3D surface with real shadows. Without additional information, the brain can't know which interpretation is correct.

AI CONNECTION

Computer vision faces the same inverse problem! This is why tasks like depth estimation, 3D reconstruction, and scene understanding are so hard. Deep learning has made progress by implicitly learning priors from massive datasets—similar to what the brain does through evolution and experience.

5.3 Evidence That Perception is Constructed

How do we know the brain constructs perception rather than passively receiving it? The evidence is overwhelming:

Timing Evidence

Visual processing takes about 100 milliseconds. Yet we can catch a baseball thrown at 90 mph, which moves 12 feet in that time. How? The brain predicts where the ball will be and generates a percept of the predicted position, not the actual position. We literally see the future, not the present.

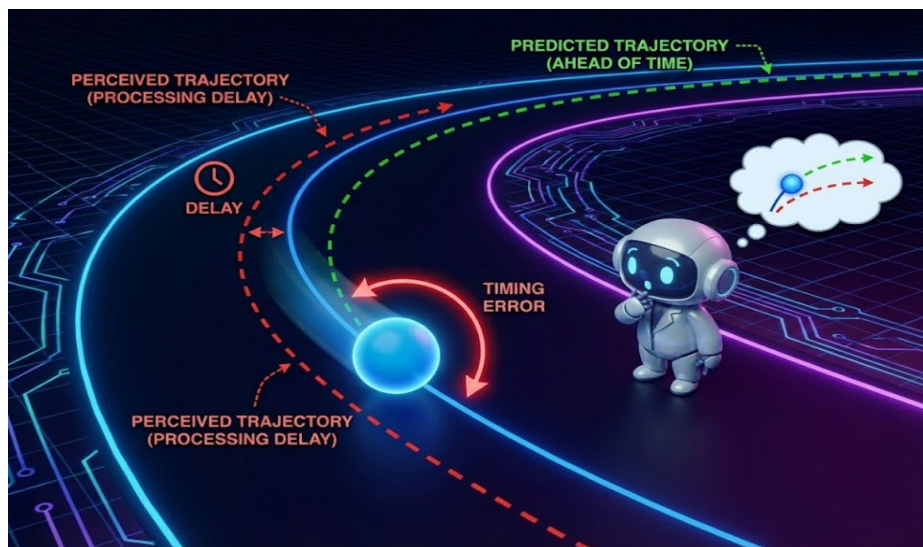


FIGURE 5.3: The Timing Problem

Blind Spot Evidence

You have a blind spot in each eye where the optic nerve exits—no photoreceptors there. Yet you don't see holes in your visual field. The brain fills in the missing information based on surrounding context. You're literally hallucinating part of your visual field all the time.



Find Your Blind Spot

1. Cover your left eye.
2. Focus your right eye on an X mark.
3. Slowly move the page closer or farther.
4. At a certain distance, a nearby O will disappear—that's your blind spot!

Notice that you don't see a hole—your brain fills in the background. This is construction, not reception.

Change Blindness

People fail to notice large changes in visual scenes when those changes occur during eye movements, blinks, or brief occlusions. If perception were passive recording, we'd notice any change. The fact that we don't proves we're not storing detailed representations—we're constructing percepts on demand.

Illusions

Visual illusions aren't failures of perception—they're the signature of a system optimized for the real world. When the brain's assumptions are violated (as in carefully constructed lab stimuli), we see things that aren't there. Illusions reveal the priors the brain uses to construct reality.

All this evidence points to the same conclusion: perception is an act of construction, not reception. The brain builds a model of what's probably out there based on ambiguous sensory data and prior knowledge. Now let's explore four theories about how it accomplishes this remarkable feat.

Section 6: The Thousand Brains Theory



CREDIBILITY CHECK

Status: SPECULATIVE but ACTIONABLE

The Thousand Brains Theory is a relatively new and still-speculative framework. It hasn't been fully validated experimentally. However, it's included here because: (1) it's directly implementable in AI systems, (2) it makes specific testable predictions, and (3) it offers a fundamentally different architecture than standard deep learning. Approach it as 'interesting hypothesis' rather than 'established fact.'

6.1 Meet Jeff Hawkins

Jeff Hawkins has had an unusual career. He founded Palm Computing (creator of the Palm Pilot), then Handspring, making him a Silicon Valley legend. But his real passion was always the brain. In 2004, he founded the Redwood Neuroscience Institute (now part of UC Berkeley) and later Numenta, a company dedicated to understanding the neocortex and applying those principles to AI.



FIGURE 6.1: Jeff Hawkins

Hawkins' 2021 book 'A Thousand Brains: A New Theory of Intelligence' presents his team's theory in accessible form. The ideas are based on over 15 years of research and are published in peer-reviewed journals, though they remain controversial in mainstream neuroscience.

6.2 The Key Insight: Cortical Uniformity

Hawkins' theory begins with a striking observation: the neocortex—the wrinkled outer layer of the brain responsible for higher cognition—has a remarkably uniform structure.

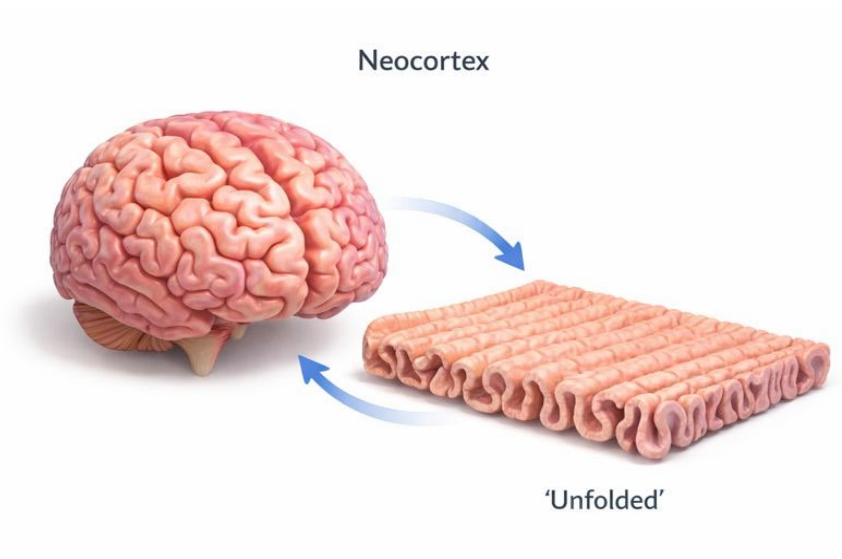


FIGURE 6.2: The Neocortex

Everywhere you look in the neocortex, you find the same basic circuit: six layers of neurons, similar connection patterns, similar cell types. Yet different regions process wildly different things—vision, hearing, touch, language, planning, math.

DID YOU KNOW?

If you could unfold and flatten the human neocortex, it would be about the size of a large dinner napkin (roughly 2,500 cm²) and about 2-3mm thick. Despite this modest size, it contains about 16 billion neurons and is responsible for everything that makes us distinctively human.

Hawkins asks: if the hardware is the same, maybe the algorithm is the same too. Maybe there's a universal algorithm that the neocortex runs everywhere, and the differences we see (vision vs. language vs. planning) are just differences in what data gets fed into the same underlying computation.

6.3 Cortical Columns: Complete World Models

The neocortex is organized into roughly 150,000 'cortical columns'—vertical structures about 2mm tall and 0.5mm wide containing roughly 100,000 neurons each.

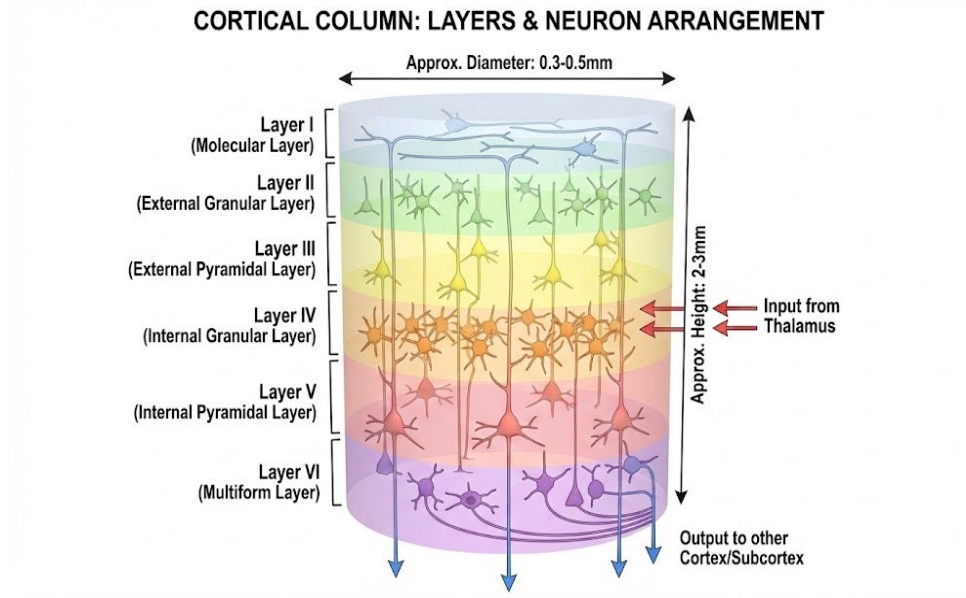


FIGURE 6.3: Cortical Column Structure

Here's Hawkins' radical proposal: each cortical column learns a complete model of whatever it's exposed to. Not a fragment, not a feature—a complete model.

This means when you see a coffee cup:

- Thousands of columns in your visual cortex each learn a complete model of 'coffee cup'
- But each column only receives input from a tiny patch of your retina
- So each column learns the object from a different perspective

This is why the theory is called 'Thousand Brains'—there are thousands of models of each object, each learned by a different cortical column from a different viewpoint.

THE THOUSAND BRAINS HYPOTHESIS

Every cortical column builds a complete model of its inputs. There are thousands of columns processing any given object, each from a different perspective. Recognition happens through a voting process among columns. The brain doesn't have ONE model of each object—it has THOUSANDS.

6.4 Reference Frames: The Structure of Knowledge

How can a column build a complete model from just a small patch of input? Hawkins' answer: reference frames.

A reference frame is a coordinate system that specifies locations relative to some anchor point. When you know an object, you don't just know its features—you know where each feature is located relative to other features.

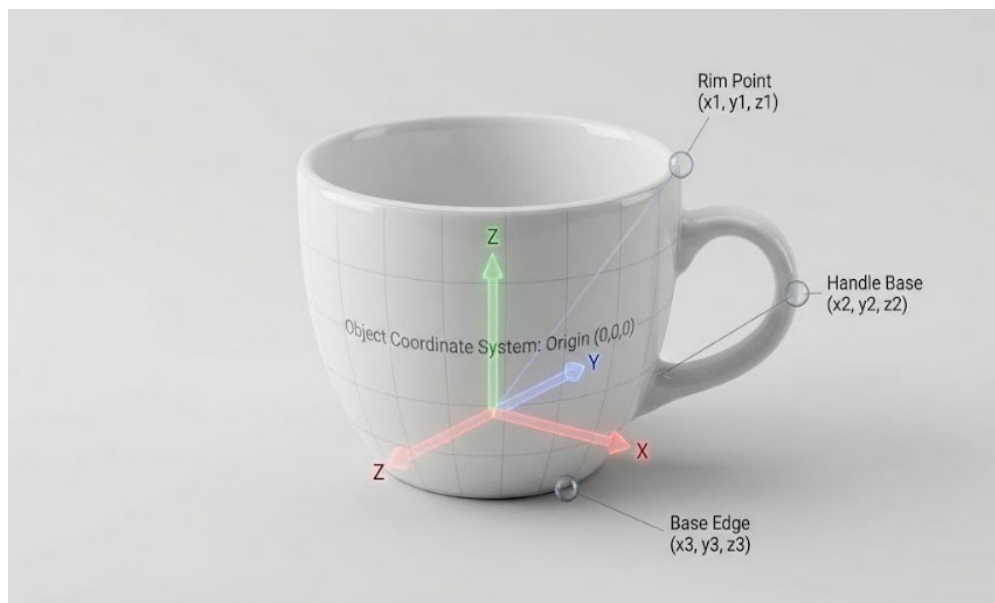


FIGURE 6.4: Reference Frames

Consider how you know a coffee cup:

- The handle is on the side
- The rim is at the top
- The base is at the bottom
- The interior is hollow

This spatial knowledge persists regardless of the cup's orientation. Even if the cup is upside down, you know the handle is still on the side—it's just that 'side' is now oriented differently relative to gravity.

The Neuroscience Foundation: Grid Cells and Place Cells

The reference frame concept isn't just theoretical speculation—it's grounded in Nobel Prize-winning neuroscience discoveries:

Discovery	Researchers	Year	Significance
Place Cells	John O'Keefe	1971 (Nobel 2014)	Neurons in hippocampus that fire when animal is at specific locations
Grid Cells	May-Britt & Edvard Moser	2005 (Nobel 2014)	Neurons in entorhinal cortex that create hexagonal coordinate grids

Grid cells create literal coordinate systems that the brain uses for spatial navigation. They fire in hexagonal patterns as you move through space, effectively giving the brain a built-in GPS system.

Hawkins' key insight: If the neocortex uses a universal algorithm, it probably uses the same reference frame mechanism for ALL knowledge, not just physical space. Abstract concepts like 'democracy' or 'recursion' might be represented using spatial-like reference frames.

AI CONNECTION

Most neural networks don't have explicit reference frames. A CNN trained on upright faces struggles with upside-down faces—it lacks the spatial structure to represent 'this is the same face, just rotated.' Hawkins argues that adding reference frames could give AI systems more human-like object understanding and generalization. This connects to recent work on equivariant neural networks and geometric deep learning.

6.5 The Voting Mechanism

If you have thousands of columns each building their own model, how do they agree on what you're perceiving? Through voting.

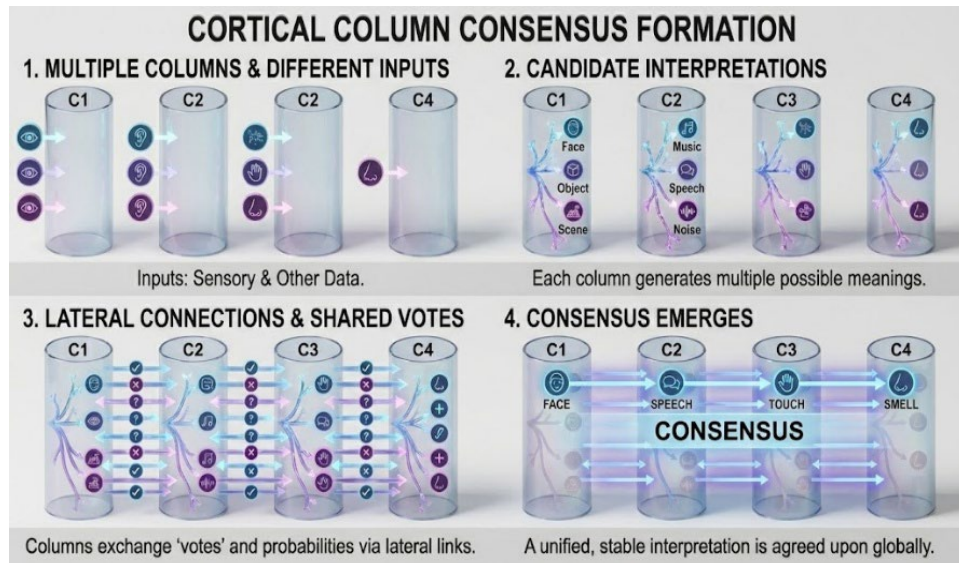


FIGURE 6.5: The Voting Mechanism

Here's how it works:

5. Each column processes its local input and generates candidate interpretations
6. Columns communicate with each other via lateral connections
7. They share their votes for what object is present
8. Consensus emerges—the interpretation that's consistent with the most columns wins
9. This consensus becomes your perception

Experience the Voting Process

Look at the famous duck-rabbit illusion. At first, you might see a duck. Then suddenly, it 'flips' to a rabbit. What's happening? Different columns are voting for different interpretations, and the balance shifts. Neither interpretation changes the raw input—what changes is which model wins the vote.

6.6 Strengths and Criticisms

Strengths

- Explains cortical uniformity with a single algorithm
- Makes specific, testable predictions about neural representations
- Directly implementable in AI (Numenta has working code)
- Addresses problems with current deep learning (viewpoint invariance, compositional knowledge)

Criticisms

- Not yet validated by mainstream neuroscience experiments
- The claim that every column has a complete model is extraordinary
- Some neuroscientists argue the cortex is more heterogeneous than Hawkins claims
- AI implementations haven't yet matched deep learning performance on standard benchmarks

Section 7: Predictive Coding & The Free Energy Principle

CREDIBILITY CHECK

Status: MAINSTREAM (Predictive Coding) / INFLUENTIAL but DEBATED (Free Energy Principle)

Predictive coding has substantial experimental support and is widely accepted in neuroscience. The Free Energy Principle is more theoretical and controversial—some see it as a unifying framework, others as too abstract to be useful. Both are important to understand.

7.1 The Core Idea: The Brain as Prediction Machine

Predictive coding proposes that the brain's primary function is not to process sensory input, but to predict it.

Here's the key insight: instead of asking 'What am I seeing?' the brain asks 'Was I right about what I expected to see?'

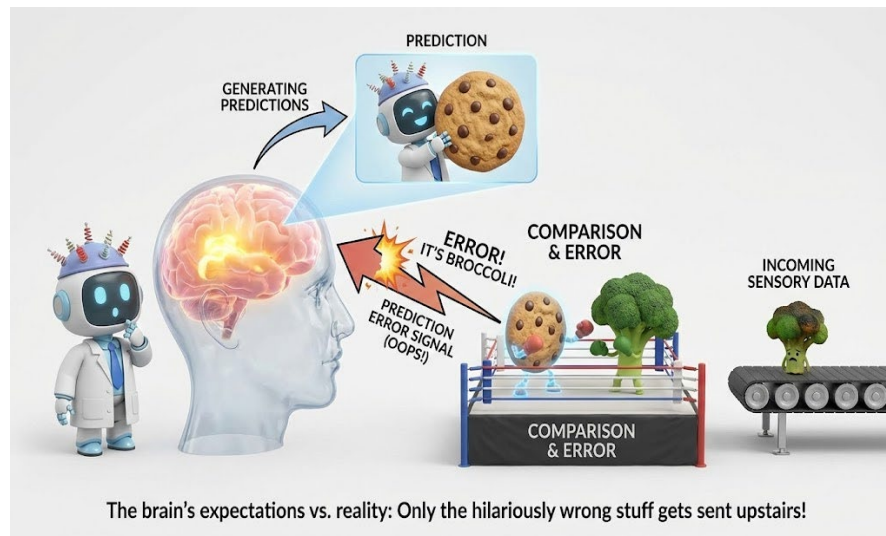


FIGURE 7.1: Predictive Coding Overview

The brain maintains a model of the world and constantly generates predictions about what sensory input should arrive. When input matches predictions, very little information needs to be processed—only the prediction errors (mismatches) get transmitted up the hierarchy.

PREDICTION ERROR MINIMIZATION

The brain doesn't transmit raw sensory data up the hierarchy. It transmits prediction errors—the difference between what was expected and what was received. Higher levels send predictions down; lower levels send errors up. Perception happens when errors are minimized—when the model's predictions match reality.

7.2 The Architecture

Predictive coding proposes a hierarchical architecture with bidirectional information flow:

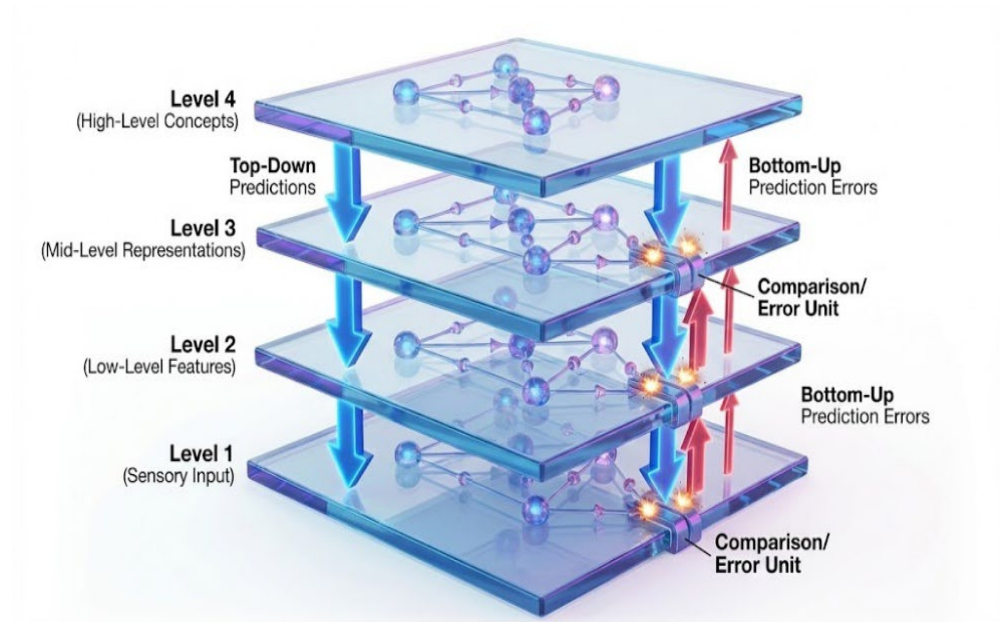


FIGURE 7.2: Predictive Coding Architecture

Top-Down: Predictions

Higher levels of the hierarchy send predictions down to lower levels. These predictions represent 'what I expect to see given my current model.' For example, if you expect to see a face, high-level 'face' neurons send predictions about where eyes, nose, and mouth should be.

Bottom-Up: Prediction Errors

Lower levels compare incoming sensory data to the predictions from above. Only the mismatch—the prediction error—gets transmitted upward. If the prediction was accurate, there's little to transmit. If it was wrong, a strong error signal propagates up.

The Update Process

When prediction errors reach higher levels, the model updates to reduce future errors. This could mean revising predictions ('oh, that's not a face, it's a mask') or updating the world model itself ('faces can look different than I thought').

AI CONNECTION

Predictive coding has interesting parallels to modern AI. Autoencoders learn to predict their own input. Variational autoencoders explicitly minimize a prediction error. Some researchers argue that transformer attention mechanisms implement something like predictive coding. The connection between predictive coding and backpropagation is an active area of research.

7.3 The Free Energy Principle

Neuroscientist Karl Friston has proposed an even more ambitious framework: the Free Energy Principle (FEP). This attempts to explain all brain function—perception, action, learning—as minimizing a single quantity called 'free energy.'



FIGURE 7.3: Karl Friston

The mathematical details are complex, but the intuition is:

- Free energy is a measure of surprise—how much the world differs from expectations
- Living systems must minimize surprise to survive (too many surprises = death)
- There are two ways to minimize surprise: change your predictions (perception) or change the world (action)

QUOTABLE

"The brain is essentially a prediction machine, and its primary function is to minimize surprise."

— Karl Friston

7.4 Active Inference: Action as Prediction

The Free Energy Principle leads naturally to 'active inference'—the idea that action and perception are two sides of the same coin.

If you want to minimize prediction error, you have two options:

- 10. **Perceptual Inference:** Update your predictions to match the world
- 11. **Active Inference:** Act on the world to make it match your predictions

A Concrete Example

Imagine a robot exploring an unfamiliar object:

- The robot touches one point and predicts: 'If this is a cube, there should be a flat surface to the left'
- It then ACTS—moves its sensor to the left to test this prediction
- If it finds a flat surface, prediction error is low → model confirmed
- If it finds a curve, high prediction error → update model (maybe it's a cylinder)

In this framework, action IS a form of inference—you gather information by acting on the world, not just by passively receiving input.

Why Active Inference Matters for AI

Traditional AI	Active Inference AI
Passive perception then action	Perception and action unified
Explore randomly or follow fixed policy	Explore to reduce uncertainty
Learn from data, then deploy	Continuous learning through interaction
Often struggles with novel situations	Naturally curious, seeks information

Active inference provides a principled framework for building agents that actively explore their environment, balance exploration and exploitation, and continuously update their world models.

RESEARCH SPOTLIGHT: PyMDP

PyMDP is an open-source Python package that implements active inference for decision-making and learning. It allows researchers and students to build agents that plan and act using predictive coding principles. This is one way to directly implement the neuroscience theories we're learning.

Link: github.com/infer-actively/pymdp

Section 8: The Bayesian Brain Hypothesis

CREDIBILITY CHECK

Status: WELL-ESTABLISHED FRAMEWORK

The Bayesian approach to perception and cognition is well-supported by experimental evidence and widely accepted. It provides a mathematical framework that makes precise predictions. The debate is more about implementation details than the basic framework.

8.1 Bayes' Rule: The Foundation

The Bayesian brain hypothesis proposes that the brain performs inference using Bayes' rule—a mathematical theorem for updating beliefs based on evidence.

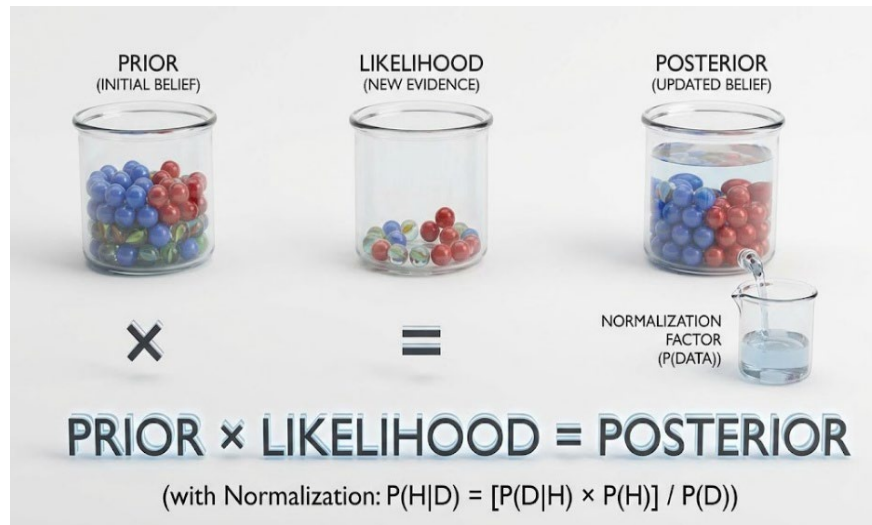


FIGURE 8.1: Bayes' Rule Explained

Here's Bayes' rule in words:

$$\text{Posterior} = (\text{Likelihood} \times \text{Prior}) / \text{Evidence}$$

Or in more intuitive terms:

- **Posterior:** What I believe after seeing the evidence
- **Likelihood:** How probable is this evidence if my hypothesis is true?
- **Prior:** What I believed before seeing the evidence
- **Evidence:** A normalizing factor

BAYESIAN INFERENCE

The brain combines prior beliefs (what you expected) with sensory evidence (what you observed) to form perceptions (what you believe). When evidence is weak or ambiguous, priors dominate. When evidence is strong and clear, it overrides priors. This explains why context and expectations so powerfully shape perception.

8.2 Priors: What You Bring to Perception

'Priors' are your pre-existing beliefs or expectations. They come from multiple sources:

- **Evolutionary priors:** Built into the brain by evolution (e.g., 'light comes from above')
- **Developmental priors:** Learned during development (e.g., 'faces have two eyes')
- **Contextual priors:** Based on current context (e.g., 'in a kitchen, round things are often plates')
- **Short-term priors:** Based on what just happened (e.g., 'the next word is probably related to the last one')

8.3 Illusions as Windows into Priors

Visual illusions reveal the priors your brain uses. Here are some classic examples:

The Light-from-Above Prior

Your brain assumes light typically comes from above (the sun, ceiling lights). This affects how you perceive 3D shape from shading. Bumps lit from above look convex; the same image flipped looks concave.

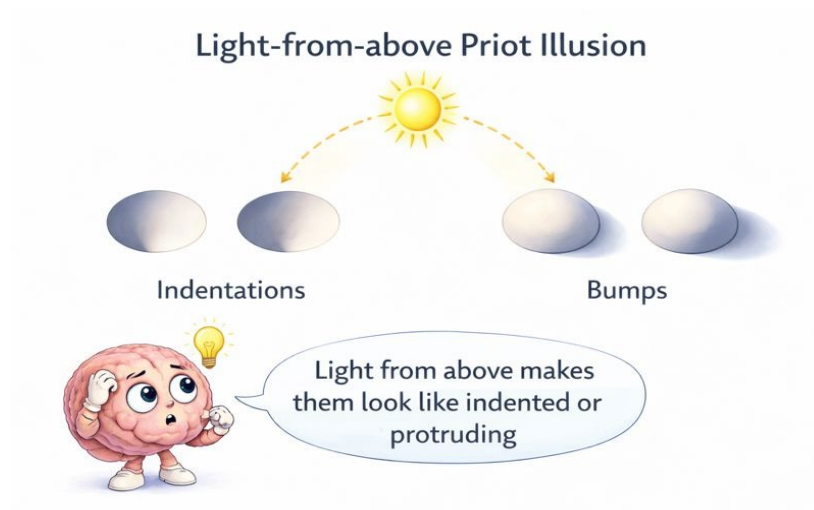


FIGURE 8.2: Light-from-Above Prior

The Size-Distance Prior

Your brain uses apparent size to infer distance. This leads to illusions like the Müller-Lyer illusion, where two lines of equal length appear different because the arrowhead endings suggest different depths.

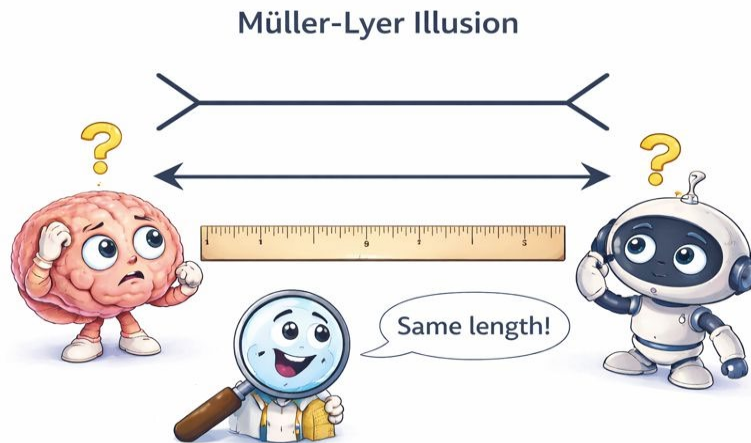


FIGURE 8.3: Müller-Lyer Illusion

AI CONNECTION

Modern AI systems often lack the structured priors that make human perception robust. A Bayesian approach suggests we should explicitly build priors into AI systems—not just train them on data and hope they learn appropriate biases. This is an active research direction in robust and few-shot learning.

Section 9: Global Workspace Theory

⚠ CREDIBILITY CHECK

Status: LEADING THEORY OF CONSCIOUSNESS (Supported but Debated)

GWT is one of the most scientifically rigorous theories of consciousness with substantial experimental support. However, a 2025 adversarial collaboration found that while GWT made correct predictions about some phenomena, it couldn't be definitively distinguished from competitor theories. Science in progress!

9.1 The Theater Metaphor

Bernard Baars originally proposed GWT in 1988 using a theater metaphor:



FIGURE 9.1: The Theater Metaphor

- **The spotlight:** Attention—what gets illuminated becomes conscious
- **The stage:** The global workspace—limited capacity for conscious content
- **The audience:** Unconscious specialized processors that receive the broadcast
- **Backstage:** Executive processes that direct the spotlight

9.2 The Neural Implementation

Stanislas Dehaene and colleagues developed a neural implementation of GWT called Global Neuronal Workspace Theory. Key features:

- Specialized modules process information unconsciously
- A network of 'workspace neurons' with long-range connections spans the cortex
- Information becomes conscious when it triggers sustained activity in the workspace
- This creates a 'global ignition'—a sudden, brain-wide broadcast

GLOBAL IGNITION

GWT predicts that the transition from unconscious to conscious processing is sudden, not gradual. When information wins the competition for workspace access, it triggers a 'global ignition'—a rapid, all-or-nothing broadcast across the brain. This has been confirmed by neuroimaging studies showing nonlinear jumps in brain activity at the threshold of awareness.

9.3 The 2025 Adversarial Collaboration

In 2025, a landmark 'adversarial collaboration' (called COGITATE) pitted GWT against a competing theory (Integrated Information Theory) in a preregistered empirical test. The results:

- Both theories made some correct predictions
- Neither theory was definitively supported or refuted
- The study highlighted how hard it is to test theories of consciousness
- Science is an ongoing process!

AI CONNECTION

GWT suggests that consciousness might require a specific architecture: specialized unconscious processors feeding into a limited-capacity global workspace that broadcasts to all modules. This is quite different from current neural networks. Some researchers are exploring 'global workspace' architectures for AI, but it remains unclear whether this would produce anything like consciousness.

Section 10: Comparing the Theories

We've now explored four major theories of brain function. Let's compare them systematically.

10.1 Summary Table

Theory	Core Claim	Focus	Status
Thousand Brains	Thousands of models with reference frames	Representation	Speculative
Predictive Coding	Brain minimizes prediction errors	Processing	Mainstream
Bayesian Brain	Perception combines priors and evidence	Computation	Well-established
Global Workspace	Consciousness = global broadcast	Access/Awareness	Leading theory

10.2 Complementary, Not Competing

These theories aren't necessarily competing—they may describe different aspects of the same system:

- **Bayesian Brain** describes WHAT computation the brain performs (probabilistic inference)
- **Predictive Coding** describes HOW that computation is implemented (hierarchical prediction errors)
- **Thousand Brains** describes the STRUCTURE of knowledge (reference frames in columns)
- **Global Workspace** describes how information becomes CONSCIOUS (broadcast to the workspace)

It's possible—even likely—that all four capture important truths about different aspects of brain function.

10.3 How They Might Work Together

Here's an example of how all four theories might operate simultaneously when you perceive a coffee cup:

INTEGRATED PROCESSING EXAMPLE

Step 1 (Bayesian): Your brain combines prior beliefs ('I'm in a kitchen, there are often cups here') with sensory evidence (visual input of a cylindrical object).

Step 2 (Predictive Coding): High-level 'cup' representations send predictions down: 'expect a handle on the side, a circular rim at top.' Lower levels compare these predictions to actual input, sending up minimal errors because predictions match.

Step 3 (Thousand Brains): Thousands of cortical columns each process their patch of input. Each column uses reference frames to encode: 'handle is at 3 o'clock relative to center.' Columns vote and reach consensus: 'this is a coffee cup.'

Step 4 (Global Workspace): The consensus representation wins competition for workspace access. Global ignition occurs—'coffee cup' is broadcast brain-wide. You become consciously aware: 'There's my cup!'

This integration isn't proven—it's a hypothesis about how these frameworks might fit together. But it illustrates that they're not necessarily in conflict.

Section 11: Neuromyths — What the Brain Doesn't Do

Before we proceed, we need to address some persistent myths about the brain. These 'neuromyths' are widely believed—even among educated people—but they're false. Understanding what the brain DOESN'T do is as important as understanding what it does.

11.1 Myth: 'We Only Use 10% of Our Brain'

The Myth: Humans only use 10% of their brain capacity. If we could access the other 90%, we'd have superhuman abilities.



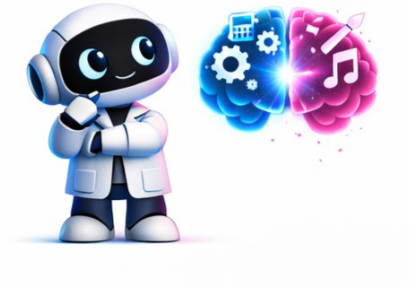
The Reality: This is completely false. Brain imaging shows that we use virtually all of our brain, and all parts have known functions. The myth may have originated from misunderstanding glial cells or from self-improvement marketing.

BY THE NUMBERS

Brain imaging studies show that over a 24-hour period, virtually 100% of the brain shows activity. Even during simple tasks, much more than 10% is active. The brain is metabolically expensive (20% of body's energy for 2% of body weight)—evolution wouldn't maintain unused tissue.

11.2 Myth: 'Left Brain = Logical, Right Brain = Creative'

The Myth: The left hemisphere is logical and analytical; the right is creative and intuitive. People are either 'left-brained' or 'right-brained.'



The Reality: While there IS hemispheric specialization (language is typically left-lateralized), the dichotomy is vastly oversimplified. Complex tasks engage both hemispheres, and the idea of 'left-brained' vs. 'right-brained' people has no scientific support.

11.3 Myth: 'Learning Styles'

The Myth: People learn better when information is presented in their preferred 'learning style' (visual, auditory, kinesthetic).



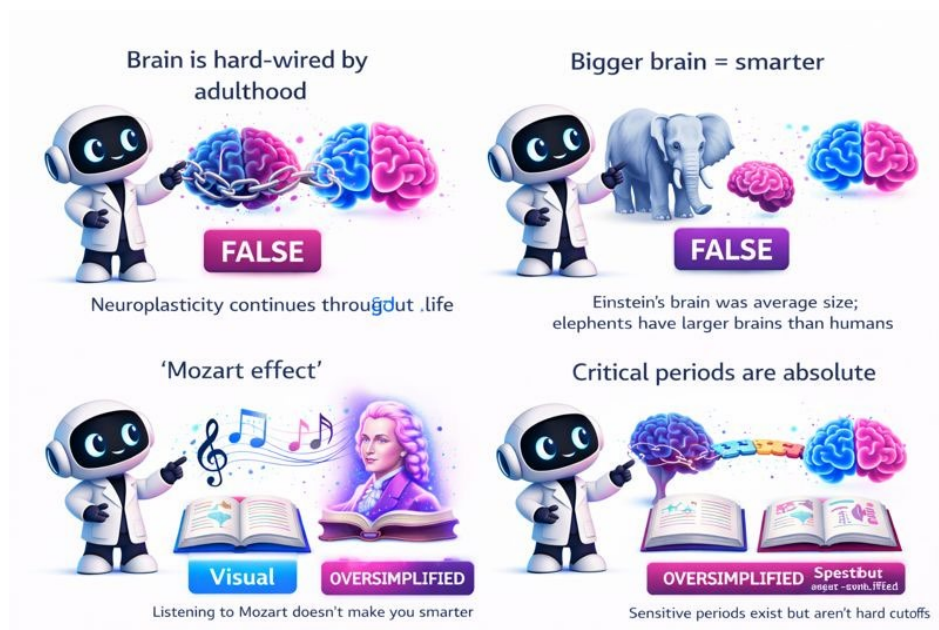
The Reality: Despite being widely taught in education programs, the learning styles hypothesis has been thoroughly debunked. Research consistently shows no benefit to matching instruction to supposed learning styles.

💡 DID YOU KNOW?

A 2012 study found that 93% of UK teachers believed in learning styles. A 2020 survey found similar numbers among US educators. Yet the research evidence is clear: there's no benefit to matching teaching to supposed learning styles. This is one of the most persistent neuromyths in education.

11.4 More Myths to Avoid

- **'Brain is hard-wired by adulthood':** FALSE—neuroplasticity continues throughout life
- **'Bigger brain = smarter':** FALSE—Einstein's brain was average size; elephants have larger brains than humans
- **'Mozart effect':** OVERSIMPLIFIED—listening to Mozart doesn't make you smarter
- **'Critical periods are absolute':** OVERSIMPLIFIED—sensitive periods exist but aren't hard cutoffs



🤖 AI CONNECTION

Neuromyths matter for AI because they can lead to misguided system designs. If you believe learning styles are real, you might waste effort building AI tutors that adapt to nonexistent categories. If you think the brain is hard-wired, you might underestimate the importance of continual learning. Critical evaluation of brain claims is essential.

Section 12: Implications for AI

What do these brain theories mean for AI development? Let's translate neuroscience insights into actionable directions.

12.1 What Current AI is Missing

Brain Principle	Current AI Gap	Potential Solution
World Models	Implicit/emergent only	Explicit world model architectures
Uncertainty	Often overconfident	Bayesian deep learning
Reference Frames	Viewpoint-dependent	Capsule networks, geometric deep learning
Prediction	Feed-forward dominant	Predictive coding networks
Global Workspace	No attention bottleneck	Workspace-style architectures
Continual Learning	Catastrophic forgetting	Brain-inspired memory consolidation

12.2 Promising Directions

World Model Architectures

Yann LeCun and others are pushing for AI systems that learn explicit world models—internal simulations that can be queried, manipulated, and used for planning. This is directly inspired by brain function.

Neuromorphic Computing

Hardware designed to mimic brain architecture: sparse, event-driven, low-power. Intel's Loihi and IBM's TrueNorth chips implement spiking neural networks.

Predictive Coding Networks

Neural networks organized hierarchically with top-down predictions and bottom-up errors. Shows promise for unsupervised learning and robustness.

BRIGHT SPOT

The connection between predictive coding and backpropagation is an active research area. Recent work shows that predictive coding can approximate backprop in biologically plausible ways. This could lead to more efficient training algorithms and more robust networks.

12.3 Concrete AI Implementations to Explore

Here are specific modern AI systems that implement brain-inspired principles—good topics for further research:

Brain Principle	AI Implementation	Key Idea	Status
World Models	DreamerV3 (DeepMind)	Learns world model for planning in imagination	State-of-art in model-based RL
World Models	JEPA (Yann LeCun)	Joint-Embedding Predictive Architecture	Active research at Meta
Predictive Coding	PredNet	Top-down predictions, bottom-up errors	Research prototype

Reference Frames	Capsule Networks (Hinton)	Equivariant representations	Research, limited adoption
Reference Frames	Slot Attention	Object-centric representations	Promising recent work
Uncertainty	MC Dropout	Approximate Bayesian inference	Widely used
Uncertainty	Bayesian Neural Networks	Explicit weight distributions	Computationally expensive
Active Inference	PyMDP	Decision-making via free energy	Open-source toolkit

These implementations show that brain-inspired AI isn't just theoretical—researchers are actively building systems based on these principles.

Section 13: Activities & Demonstrations

This section contains hands-on activities to deepen your understanding of the concepts covered in this module.

13.1 Visual Illusions Demonstration

Illusions aren't failures of perception—they're windows into how your brain constructs reality. Each illusion reveals specific priors your brain uses.

Activity: Checker Shadow Illusion

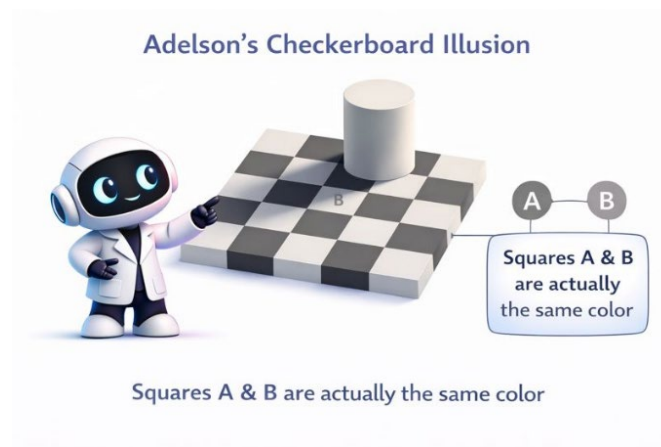


FIGURE 13.1: Checker Shadow Illusion

In this famous illusion by Edward Adelson, squares A and B are identical in brightness (check it by covering the surrounding context). Your brain 'corrects' for the assumed shadow, making B appear lighter.

What it reveals: Light-from-above prior, surface constancy, context effects

Activity: 'The Dress'

In 2015, a photo of a dress went viral because people disagreed on its color—some saw white and gold, others blue and black



What it reveals: Color constancy priors differ between individuals based on assumed lighting conditions. The dress real color? Black and blue

13.2 Which Theory Explains This?

For each scenario below, discuss which theory (or theories) best explains the phenomenon:

Scenario 1: The Cocktail Party Effect

You're at a noisy party, focused on a conversation. You're not consciously processing other conversations. Then someone across the room says your name, and your attention instantly shifts.

DISCUSSION

Which theory best explains this? Consider: What was being processed unconsciously? What triggered the attention shift? What 'won' access to awareness?

Scenario 2: Robot Grasping a Novel Object

A robot trained on thousands of objects encounters something it's never seen before—a strangely-shaped art piece. A human could pick it up easily; the robot struggles.

DISCUSSION

Which brain principles is the robot missing? Consider: How do humans represent objects? How do we generalize from known to unknown?

Scenario 3: LLM Hallucination

A large language model confidently states that a made-up historical event actually occurred. It provides detailed 'facts' about something that never happened.

DISCUSSION

Which brain function is the LLM missing? Consider: How do humans know what they don't know? What role do world models play in detecting falsehoods?

13.3 Design Challenge

Apply brain principles to improve an AI system:

The Task: Design a vision system for a household robot that needs to recognize objects from any angle, in various lighting conditions, and including objects it's never seen before.

Consider incorporating:

- Reference frames (from Thousand Brains)
- Predictive processing
- Explicit priors (from Bayesian Brain)
- Uncertainty representation



Section 14: Lab Preview — Cortical Column Simulation

In the accompanying lab, you'll implement a simplified simulation of cortical column voting based on the Thousand Brains Theory. Detailed instructions will be provided in a separate document

14.1 What You'll Build

- Multiple 'columns' that each receive partial sensory input
- Each column maintains candidate object hypotheses
- A voting mechanism for columns to reach consensus
- Visualization of how consensus emerges over time

14.2 Learning Goals

- Understand how distributed representations can be combined through voting
- Experience the trade-off between speed and accuracy in consensus mechanisms
- Connect abstract theory to concrete implementation
- Identify limitations of simplified models

14.3 Algorithm Overview

The simulation follows these steps:

12. Initialize N columns, each with access to a different 'patch' of input
13. Each column generates initial hypotheses based on its local input
14. Columns share votes with neighbors (simulating lateral connections)
15. Each column updates its beliefs based on received votes
16. Repeat until consensus emerges or maximum iterations reached
17. Output: the winning hypothesis and confidence level

14.4 Technical Details

- **Platform:** Python (with NumPy) or Wolfram Language
- **Estimated time:** 2-3 hours
- **Prerequisites:** Basic Python or Wolfram programming, understanding of this module's content
- **Starter code:** Available in Canvas under Module 02 Lab Materials

Section 15: Chapter Summary

The Central Insight

The brain constructs reality rather than passively receiving it. What you experience as 'the world' is actually an internal model—a simulation—built from ambiguous sensory data and prior knowledge.

Four Theories of How

18. **Thousand Brains Theory:** Thousands of cortical columns each learn complete models using reference frames; they reach consensus through voting. (Speculative but actionable for AI)
19. **Predictive Coding:** The brain generates predictions and transmits only prediction errors. Perception is minimizing surprise. (Mainstream neuroscience)
20. **Bayesian Brain:** The brain combines prior beliefs with sensory evidence using Bayes' rule. (Well-established framework)
21. **Global Workspace Theory:** Consciousness arises when information wins competition for access to a global workspace and gets broadcast brain-wide. (Leading consciousness theory)

Neuromyths to Avoid

- We don't just use 10% of our brain (we use it all)
- The left-brain/right-brain dichotomy is oversimplified
- Learning styles have no scientific support
- The brain isn't hard-wired—neuroplasticity continues throughout life

Implications for AI

Brain principles suggest current AI is missing: explicit world models, proper uncertainty handling, reference frames for viewpoint-invariant knowledge, predictive processing, and mechanisms for knowing what it doesn't know.

Looking Ahead: Module 03

In Module 03, we'll explore the physical structures that implement these theories. You'll learn:

- How the prefrontal cortex might serve as the global workspace
- Where cortical columns actually live in the brain
- Which brain regions generate the predictions described by predictive coding
- The role of the thalamus as a relay and integration hub
- How neural plasticity enables learning and adaptation

The theoretical frameworks you learned in Module 02 will become concrete as we map them onto actual brain anatomy.

Section 16: Review Questions

Knowledge Check

1. What is the 'inverse problem' of perception, and why is it hard?
2. Explain the key difference between passive reception and active construction of perception.
3. What is a 'reference frame' in the Thousand Brains Theory, and why does it matter?
4. In predictive coding, what gets transmitted up the hierarchy—predictions or prediction errors? Why?
5. What is a 'prior' in Bayesian terms, and give an example of an evolutionary prior.
6. Explain the 'global ignition' concept from Global Workspace Theory.
7. Why is the '10% of the brain' claim a myth? What evidence disproves it?
8. Name two ways brain principles suggest we could improve current AI systems.
9. What is the difference between sensation and perception?
10. How do grid cells relate to the concept of reference frames in the Thousand Brains Theory?

Discussion Questions

11. If all knowledge requires reference frames (as Thousand Brains claims), what does this mean for LLMs that process text without spatial grounding?
12. Predictive coding suggests perception is mostly about confirming expectations. How might this change how we design AI perception systems?
13. The 2025 COGITATE study couldn't definitively choose between consciousness theories. What does this tell us about the difficulty of studying the brain?
14. Why do neuromyths persist even among educated people? How can understanding this help us be better AI developers?
15. Compare and contrast perceptual inference and active inference. Give an example of each in the context of a robot exploring a new environment.

Section 17: Glossary

Active Inference

Extension of predictive coding where agents minimize prediction error by either updating beliefs OR acting on the world to make predictions come true.

Approximate Bayesian Inference

Methods for approximating Bayesian computations when exact calculation is intractable, as the brain must do.

Backpropagation

Algorithm for training neural networks by propagating errors backward. Debated whether the brain uses something similar.

Bayes' Rule

Mathematical theorem for updating beliefs: $\text{Posterior} = (\text{Likelihood} \times \text{Prior}) / \text{Evidence}$. Foundation of the Bayesian Brain hypothesis.

Bayesian Brain Hypothesis

Theory that the brain represents uncertainty and performs probabilistic inference using principles akin to Bayes' rule.

Blind Spot

Area of the retina with no photoreceptors (where optic nerve exits). The brain fills in this gap—evidence of constructive perception.

Change Blindness

Failure to notice large changes in visual scenes during disruptions. Evidence that we don't store detailed representations.

Cortical Column

Vertical structure in neocortex (~0.5mm wide, 2mm tall) containing ~100,000 neurons. May implement a universal algorithm.

Counterfactual Reasoning

The ability to reason about 'what if' scenarios—what would happen if something were different. Requires a manipulable world model.

Embodied Cognition

The theory that cognitive processes are deeply rooted in the body's interactions with the world, not just abstract computation in the brain.

Free Energy

In Friston's framework, an information-theoretic quantity that bounds surprise. Minimizing free energy = minimizing prediction error.

Free Energy Principle

Karl Friston's proposal that all brain function can be understood as minimizing a quantity called free energy (related to surprise).

Global Ignition

In Global Workspace Theory, the sudden, all-or-nothing transition when information enters conscious awareness.

Global Neuronal Workspace Theory

Neural implementation of Global Workspace Theory, proposed by Dehaene and colleagues.

Global Workspace Theory

Theory that consciousness arises when information wins competition for access to a limited-capacity workspace and is broadcast brain-wide.

Grid Cells

Neurons in the entorhinal cortex that fire in hexagonal patterns during spatial navigation. Won the 2014 Nobel Prize. Foundation for reference frame concept.

Helmholtz

Hermann von Helmholtz (1821-1894), German scientist who proposed that perception involves 'unconscious inference.'

Hierarchical Temporal Memory

Computational framework developed by Numenta based on neocortical principles. Precursor to Thousand Brains Theory.

Inverse Problem

The challenge of inferring 3D scene structure from 2D retinal images. Has infinite solutions without additional constraints.

Lateral Connections

Horizontal connections between neurons in the same layer. May enable voting between cortical columns.

Likelihood

In Bayesian terms, the probability of observing the evidence given a hypothesis.
 $P(\text{Evidence}|\text{Hypothesis})$.

Neocortex

The outer layer of the cerebral cortex, responsible for higher cognitive functions. Remarkably uniform in structure.

Neuroethics

The field that examines ethical, legal, and social implications of neuroscience research and its applications.

Neuroplasticity

The brain's ability to change and reorganize throughout life. Contradicts the myth that the brain is 'hard-wired.'

Neuromyth

A widely-believed but false claim about how the brain works. Examples: '10% of brain,' 'learning styles.'

Perception

The brain's constructed interpretation of sensory signals. Active inference, can be 'wrong' (illusions).

Place Cells

Neurons in the hippocampus that fire when an animal is at specific locations. Discovered by O'Keefe (Nobel 2014).

Posterior

In Bayesian terms, the updated belief after considering evidence. What we believe after combining prior and likelihood.

Prediction Error

The mismatch between what was predicted and what was observed. The core signal in predictive coding.

Predictive Coding

Theory that the brain generates predictions and transmits only prediction errors up the processing hierarchy.

Prior

In Bayesian terms, the belief held before seeing evidence. Can come from evolution, development, or context.

Reference Frame

A coordinate system for representing location. In Thousand Brains, knowledge is structured relative to reference frames.

Sensation

The raw transduction of physical energy into neural signals. Relatively faithful to physical reality.

Sparse Coding

Neural coding scheme where only a small fraction of neurons are active at once. Efficient and may enable pattern separation.

Thousand Brains Theory

Jeff Hawkins' theory that the neocortex consists of thousands of columns, each building complete world models using reference frames.

Top-Down Processing

Information flow from higher to lower levels of the processing hierarchy. Carries predictions in predictive coding.

Unconscious Inference

Helmholtz's term for the automatic, below-awareness inferences the brain makes to interpret sensory data.

Voting Mechanism

In Thousand Brains, the process by which columns reach consensus about the identity of perceived objects.

World Model

An internal representation of external reality that supports prediction, imagination, and planning.

Section 18: References & Further Reading

Primary Sources

Thousand Brains Theory

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Global Workspace Theory

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- Baars, B. J. (1988). *A Cognitive Theory of Consciousness*. Cambridge University Press.
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Grid Cells & Place Cells

- O'Keefe, J., & Nadel, L. (1978). *The Hippocampus as a Cognitive Map*. Oxford University Press.
- Moser, E. I., Kropff, E., & Moser, M. B. (2008). Place cells, grid cells, and the brain's spatial representation system. *Annual Review of Neuroscience*.

Accessible Books

- Seth, A. (2021). Being You: A New Science of Consciousness. Dutton.
- Eagleman, D. (2015). The Brain: The Story of You. Pantheon.
- Barrett, L. F. (2017). How Emotions Are Made: The Secret Life of the Brain. Houghton Mifflin Harcourt.

Online Resources

- Numenta.com — Research papers and code for Thousand Brains implementations
- HTM School (YouTube) — Video series explaining Hierarchical Temporal Memory concepts
- BrainFacts.org — Neuroscience education resources from the Society for Neuroscience
- PyMDP (github.com/infer-actively/pymdp) — Python package for active inference simulations
- DreamerV3 (github.com/danijar/dreamerv3) — World model implementation for reinforcement learning

— End of Module 02 —

[Next: Module 03 — Brain Anatomy & Functions](#)